

Optimal Plug-in Electric Vehicle Charging in Multi-Period Power Markets: A Stochastic Dynamic Programming Approach

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1 Introduction

The rapid growth of plug-in electric vehicle (PEV) adoption is reshaping how electricity is consumed at the household and grid level. A modern EV battery holds 40–100 kWh of energy, and charging it represents a significant, recurring cost for the owner. In a deregulated power market (such as ISO-NE, NYISO, and PJM), wholesale electricity prices vary substantially across hours of the day and also across days. Hence, an owner wishing to charge their EV faces an important problem. When should the battery be charged to minimize expected cost while ensuring sufficient energy is available at departure?

The above problem is sequential and stochastic in nature, as the owner must commit to each period’s energy purchase before the corresponding market price is known. Charging too aggressively at the wrong moment locks an unfavorable price, while waiting too long risks paying a steep penalty for arriving at a departure with an almost empty battery. The framework for such problems is a stochastic dynamic programming (DP) problem, which has been studied extensively in optimal-control and energy-systems literature [1].

A second source of complexity, often glossed over in early formulations, is the model of the battery itself. A real lithium-ion cell is not perfect. As charging losses grow with current, charging becomes harder to access at high rates (the rate-capacity effect), and resting after heavy use temporarily restores apparent capacity (the recovery effect). The Kinetic Battery Model (KiBaM), introduced by Manwell and McGowan [2], captures these phenomena via a two-compartment diffusion model that has become a standard for battery-aware modeling.

This project formulates and numerically solves the optimal PEV charging problem in a multi-period power market under two battery representations. The first treats the battery as ideal and yields the simplest possible DP. The second uses KiBaM, which introduces a richer state space and qualitatively different policy structure. Both problems are solved by backward induction on a discrete state grid, and the resulting policies are compared.

The remainder of the paper is organized as follows. Section 2 sets up the shared notation. Section 3 develops the price model. Sections 4 and 5 formulate the ideal- and KiBaM-battery DPs respectively. Section 6 describes the numerical method. Section 7 reports results, and Section 8 concludes.

2 Notation and Preliminaries

Table 1 summarizes the notation used throughout the paper.

Table 1: Nomenclature.

Symbol	Meaning
$k \in \{0, 1, \dots, N-1\}$	Decision time index
N	Number of decision steps
Δt	Period length (hr)
$x_k \in [0, E_{\max}]$	Remaining energy demand, ideal model (kWh)
$x_{1,k} \in [0, x_{1,\max}]$	Available charge, KiBaM (kWh)
$x_{2,k} \in [0, x_{2,\max}]$	Bound charge, KiBaM (kWh)
$E_{\max}$	Total battery capacity (kWh)
$u_k \in [0, u_{\max}]$	Energy purchased in period k (kWh)
$P_{\max}$	Charger power cap (kW); $u_{\max} = P_{\max}\Delta t$
α	KiBaM rate constant (1/hr)
w_k	Spot electricity price in period k (\$/kWh)
K	Long-run mean of the price process (\$/kWh)
Δ	Half-width of the unconditional price support (\$/kWh)
g_k	Stage cost; $g_k = w_k u_k$
g_N	Terminal cost; penalty for unmet demand
K_T	Per-kWh terminal-penalty rate (\$/kWh)
s_k	Augmented state (state-dim differs by battery model)
$J_k(\cdot)$	Optimal cost-to-go function at stage k
$\mu_k^*(\cdot)$	Optimal policy at stage k
N_q	Number of Gauss–Legendre quadrature nodes
$\tilde{\omega}_i, w_i$	Quadrature weights (normalized) and nodes

3 Modeling the Electricity Price Process

3.1 Overview

The price market model captures how electricity spot prices evolve over the charging horizon. It is the only source of randomness in the entire DP problem, as the battery dynamics (both ideal and KiBaM) are deterministic. The price process therefore drives the entire stochastic structure of the problem and is the reason the dynamic program requires state augmentation.

We model the price at step k as conditionally uniform on a support that depends on the previous observed price:

$$w_k \mid w_{k-1} \sim \mathcal{U}[\max(0, 2w_{k-1} - K - \Delta), K + \Delta], \quad (1)$$

parameterized by a mean K and a half-width Δ . Throughout, we use $K = 0.30$ \$/kWh, $w_{-1} = 0.30$ \$/kWh, and a sensitivity sweep $\Delta \in \{0.05, 0.10, 0.20\}$ \$/kWh. The unconditional support of the price is $[K - \Delta, K + \Delta]$.

3.2 Martingale Property

Although the support of w_k shifts with w_{k-1} , the conditional mean is unchanged:

$$\mathbb{E}[w_k | w_{k-1}] = \frac{(2w_{k-1} - K - \Delta) + (K + \Delta)}{2} = w_{k-1}. \quad (2)$$

The price process is therefore a martingale: the current observed price is the best forecast of the future price, and there is no exploitable drift. This martingale identity holds in the interior region $w_{k-1} \geq (K + \Delta)/2$ where the lower truncation in Eq. (1) is inactive. When $w_{k-1} < (K + \Delta)/2$ the lower endpoint is clipped at zero and the conditional mean equals $(K + \Delta)/2$, producing additional mean reversion at low prices.

3.3 Mean Reversion through the Support

The support is asymmetric about w_{k-1} except at the long-run mean K . The lower endpoint moves with w_{k-1} while the upper endpoint is fixed, which produces soft mean reversion. When w_{k-1} reaches $K + \Delta$, the support collapses to a point and $w_k = w_{k-1}$ deterministically. Hence, the conditional variance,

$$\text{Var}(w_k | w_{k-1}) = \frac{1}{3}(K + \Delta - w_{k-1})^2, \quad (3)$$

is largest at low prices and vanishes at the cap.

3.4 Information Timing and State Augmentation

The control u_k is committed before w_k is observed hence, only w_{k-1} is available at decision time. Because the conditional distribution of w_k depends on w_{k-1} , the natural physical state is not Markov on its own, and we augment with the most recent price:

$$s_k = (x_k, w_{k-1}) \quad (\text{ideal}), \quad s_k = (x_{1,k}, x_{2,k}, w_{k-1}) \quad (\text{kinetic}). \quad (4)$$

3.5 Numerical Implementation

The conditional expectation in the Bellman recursion is computed by an N_q -point Gauss–Legendre rule on $[a, b] = [\max(0, 2w_{k-1} - K - \Delta), K + \Delta]$, with weights normalized so $\sum_i \tilde{\omega}_i = 1$; this preserves the martingale property (2) at the discretization level. Forward simulations draw directly from $\mathcal{U}[a, b]$. Also, the deterministic limit $\Delta \rightarrow 0$ is used as a verification check i.e. the policy should reduce to charging fully if and only if $w_0 < K_T$.

4 Part (a): Ideal Battery Model

4.1 System Dynamics

The ideal model treats the battery via the demand-depletion dynamics

$$x_{k+1} = x_k - u_k, \quad x_k \in [0, E_{\max}], \quad u_k \in [0, u_{\max}], \quad (5)$$

where x_k is the remaining energy demand and u_k is the energy purchased in period k .

4.2 Stage Cost and Terminal Penalty

The per-period cost is

$$g_k = w_k u_k. \quad (6)$$

The departure SOC constraint is softened via a terminal penalty proportional to remaining unmet demand,

$$\Phi(x_N) = K_T x_N, \quad (7)$$

with K_T chosen large enough that the constraint binds almost surely at the optimum; equivalently one may set $\Phi(x_N) = +\infty \cdot \mathbf{1}\{x_N > 0\}$ and enforce feasibility of $\mathcal{U}_k(x_k)$ directly.

4.3 Optimization Problem

$$\min_{\{\mu_k\}_{k=0}^{N-1}} \mathbb{E} \left[\sum_{k=0}^{N-1} w_k u_k + \Phi(x_N) \right], \quad u_k = \mu_k(x_k, w_{k-1}). \quad (8)$$

4.4 Bellman Equation

The optimal cost-to-go satisfies

$$J_k(x, w) = \min_{u \in \mathcal{U}(x)} \mathbb{E}_{w_k|w} [w_k u + J_{k+1}(x - u, w_k)] \quad (9)$$

with $J_N(x) = \Phi(x)$ and feasible set

$$\mathcal{U}(x) = \{u \in [0, u_{\max}] : u \leq x\}. \quad (10)$$

5 Part (b): KiBaM Battery Model

5.1 Two-Tank Representation

The KiBaM, introduced by Manwell and McGowan [2], represents the battery as two coupled charge tanks. An *available* tank $x_{1,k}$ that can be drawn from or charged directly, and a *bound* tank $x_{2,k}$ that exchanges charge with the available tank through a diffusion process. The structure is illustrated in Figure 1.

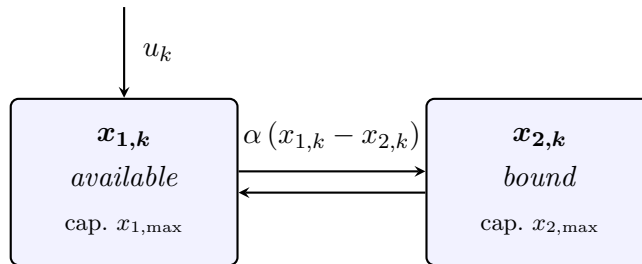


Figure 1: KiBaM two-tank representation with diffusion rate $\alpha(x_{1,k} - x_{2,k})$.

5.2 State Dynamics

A forward-Euler discretization gives

$$x_{1,k+1} = x_{1,k} + u_k - \alpha \Delta t (x_{1,k} - x_{2,k}), \quad (11)$$

$$x_{2,k+1} = x_{2,k} + \alpha \Delta t (x_{1,k} - x_{2,k}). \quad (12)$$

Adding the two equations gives $x_{1,k+1} + x_{2,k+1} = x_{1,k} + x_{2,k} + u_k$ — energy is conserved. The diffusion rate α is the only KiBaM-specific parameter.

5.3 Control Constraint

$$0 \leq u_k \leq \min(P_{\max} \Delta t, \alpha(x_{1,\max} - x_{2,k})). \quad (13)$$

The second term in the minimum produces CC-CV (Constant Current- Constant Voltage) like tapering endogenously: as the bound tank fills toward capacity, the maximum admissible charge rate shrinks to zero.

5.4 Bellman Equation

The one-period cost is the electricity payment incurred in period k :

$$g_k(x_{1,k}, x_{2,k}, w_k, u_k) = w_k u_k. \quad (14)$$

Since u_k is chosen before w_k is realized, the Bellman recursion takes an expectation over the conditional price distribution $w_k \mid w_{k-1}$. The terminal value penalizes any remaining unfilled capacity:

$$J_N(x_{1,N}, x_{2,N}) = K_T (E_{\max} - x_{1,N} - x_{2,N}), \quad E_{\max} = x_{1,\max} + x_{2,\max} \quad (15)$$

where the state constraints ensure $x_{1,N} + x_{2,N} \leq E_{\max}$.

The objective is to minimize the expected total charging cost over the horizon, consisting of the sum of stage costs and the terminal penalty:

$$\mathbb{E} \left[\sum_{k=0}^{N-1} w_k u_k + J_N(x_{1,N}, x_{2,N}) \right]. \quad (16)$$

By the principle of optimality, the optimal cost-to-go function satisfies a recursive decomposition in which the current decision trades off immediate cost against expected future cost. With augmented state $s_k = (x_{1,k}, x_{2,k}, w_{k-1})$,

$$J_k(x_1, x_2, w) = \min_{u_k} \mathbb{E}_{w_k \mid w} [w_k u_k + J_{k+1}(x_{1,k+1}, x_{2,k+1}, w_k)]. \quad (17)$$

6 Numerical Method

6.1 Discretization

- **Part (a) — ideal:** state grids $N_x = N_w = 31$, control grid $N_u = 31$, quadrature points $N_q = 5$, horizon $N = 8$.
- **Part (b) — KiBaM:** state grids $N_{x_1} = N_{x_2} = 11$, $N_w = 9$; control grid $N_u = 9$; quadrature points $N_q = 3$; horizon $N = 8$. The smaller grids reflect the higher-dimensional state space.
- **Price quadrature:** Gauss–Legendre nodes mapped to the conditional support $[\max(0, 2w_{k-1} - K - \Delta), K + \Delta]$ and normalized to a probability measure (Section 3).
- **Sensitivity sweep:** $K_T \in \{0.1, 0.2, 0.3, 0.4, 0.5, 0.7, 1.0\}$ \$/kWh and $\Delta \in \{0.05, 0.10, 0.20\}$ \$/kWh, for $7 \times 3 = 21$ runs per battery model.
- **Physical parameters:** $E_{\max} = 40$ kWh, $P_{\max} = 7$ kW, $\Delta t = 1$ hr (so $u_{\max} = 7$ kWh), $\alpha = 0.30$ /hr, $K = 0.30$ \$/kWh, $w_{-1} = 0.30$ \$/kWh, and $x_{1,\max} = x_{2,\max} = 20$ kWh.

6.2 Implementation Notes

- Language: Python 3 with NumPy.
- Off-grid value-function lookups use bilinear interpolation in Part (a) and trilinear interpolation in Part (b), implemented in pure NumPy.
- State-space cardinality per time slice: $N_x \cdot N_w = 961$ for Part (a) and $N_{x_1} \cdot N_{x_2} \cdot N_w = 1,089$ for Part (b).
- Inner DP iterations per (K_T, Δ) solve: ~ 1.2 M for Part (a) and ~ 235 k for Part (b).
- All sensitivity results, sample paths, and parameter settings are exported to a multi-sheet Excel workbook for reproducibility.

6.3 Validation Checks

1. **Deterministic limit** ($\Delta \rightarrow 0$). As the price spread vanishes, the conditional support of w_k collapses to a point and the problem becomes deterministic. The optimal policy should reduce to charging fully whenever $w_0 < K_T$ and not at all otherwise. We verify this numerically with $\Delta = 10^{-4}$.
2. **Martingale preservation by quadrature.** For every w_{k-1} grid point, $\sum_i \tilde{\omega}_i w_i = w_{k-1}$, confirming that the discretized expectation inherits the martingale property of the continuous price process.
3. **Energy conservation in KiBaM.** The forward-Euler update satisfies $x_{1,k+1} + x_{2,k+1} = x_{1,k} + x_{2,k} + u_k$ exactly, ensuring that all charge purchased ends up in one of the two tanks.
4. **Recovery-effect demonstration.** An exogenous charge–use–rest scenario produces the expected geometric decay of $x_{1,k} - x_{2,k}$ during rest, with decay ratio matching $1 - 2\alpha \Delta t$ — verifying both the model and its forward-Euler discretization.

7 Numerical Results

All results below are computed at the representative case $K_T = 0.5$ \$/kWh and $\Delta = 0.10$ \$/kWh unless otherwise noted. The full sensitivity sweep over $K_T \in \{0.1, 0.2, 0.3, 0.4, 0.5, 0.7, 1.0\}$ and $\Delta \in \{0.05, 0.10, 0.20\}$ ($7 \times 3 = 21$ runs per model) is reported in Table 3.

7.1 Part (a): Ideal Battery

Figure 2 shows the optimal policy at three time slices. The policy displays the expected bang-bang structure with a sharp price threshold near $w \approx K_T$, i.e., at every (k, x) the optimal control is essentially 0 or $u_{\max}$. Figure 3 shows a representative sample-path simulation under the optimal policy together with the value function $J_1(x, w_0)$ for three values of the initial price. The piecewise-linear shape of J_1 in x is consistent with the analytical solution at $k = N - 1$, where the slope equals $\min(w_0, K_T)$.

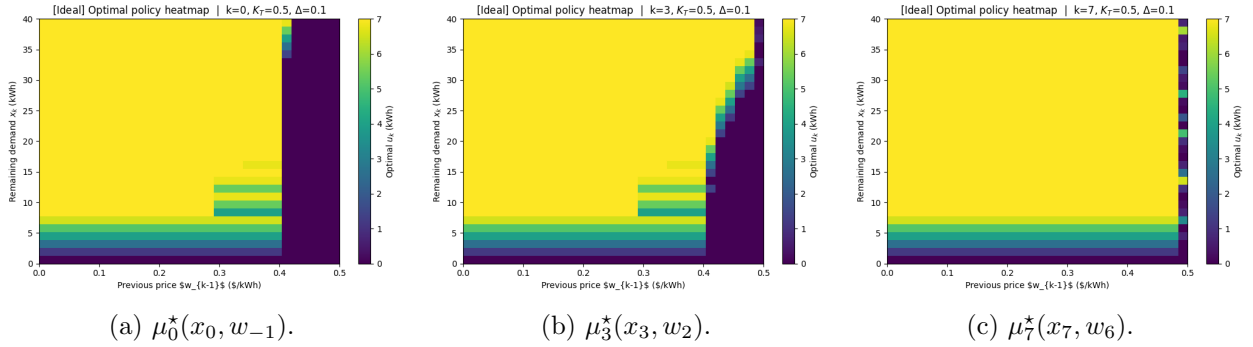
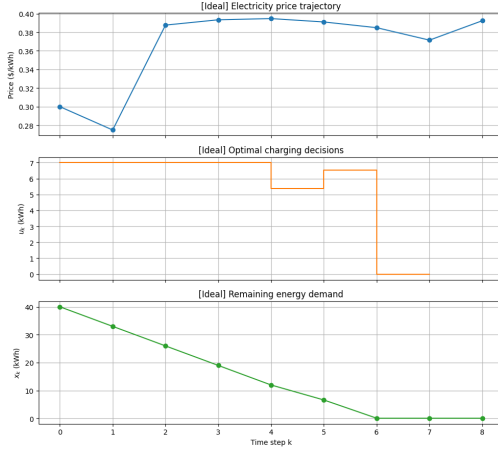
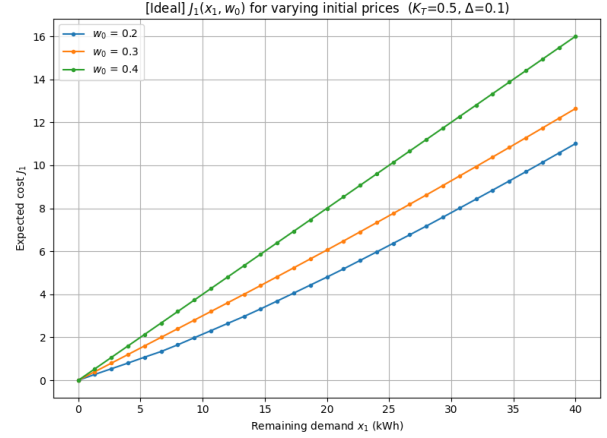


Figure 2: Ideal-battery optimal policy at three time slices. A vertical price threshold near $w \approx K_T$ separates a maximal-charge region (left) from a no-charge region (right).



(a) Sample trajectory under the optimal policy. Top: realized price path. Middle: optimal charging decisions, dominated by bang-bang behavior with two interior values at $k \in \{4, 5\}$ where the demand constraint or future option value produces a non-extreme optimum. Bottom: remaining demand x_k , reaching zero at $k = 6$ — the EV departs fully charged.



(b) $J_1(x, w_0)$ is approximately linear in x within the feasible charging window, with slopes 0.275, 0.315, 0.400 for $w_0 = 0.2, 0.3, 0.4$ respectively. These slopes are bounded above by $K_T = 0.5$ and approach w_0 as $w_0 \rightarrow K_T$, reflecting that the expected per-kWh charging cost is the mean price across the remaining horizon.

Figure 3: Trajectory and value-function diagnostics for the ideal-battery DP.

7.2 Part (b): Kinetic Battery (KiBaM)

Figure 4 shows the three-dimensional KiBaM policy at a fixed mid-grid value of x_2 , showcasing how the price threshold on the w -axis is preserved across k but the maximum admissible control shrinks compared to the ideal model. Figure 5 reports the full policy surface $u_0^*(x_1, x_2)$ alongside the value-function surface $J_1(x_1, x_2)$. The CC-CV-like taper near $x_2 \rightarrow x_{1,\max}$ is visible directly on the policy surface and arises endogenously from the KiBaM rate constraint $\alpha(x_{1,\max} - x_2)$. Figure 6 shows the closed-loop sample path together with the canonical charge–use–rest experiment that demonstrates the recovery effect.

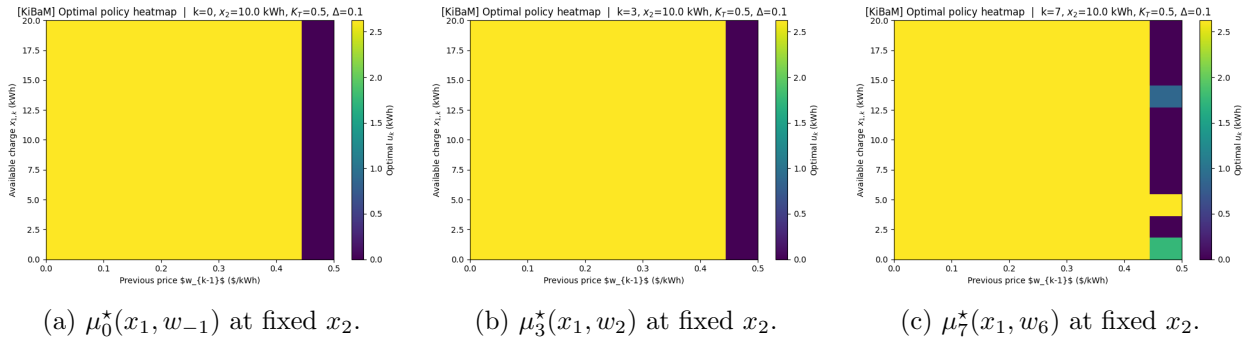


Figure 4: KiBaM optimal policy at three time slices, with the bound-tank coordinate x_2 fixed at the middle of its grid.

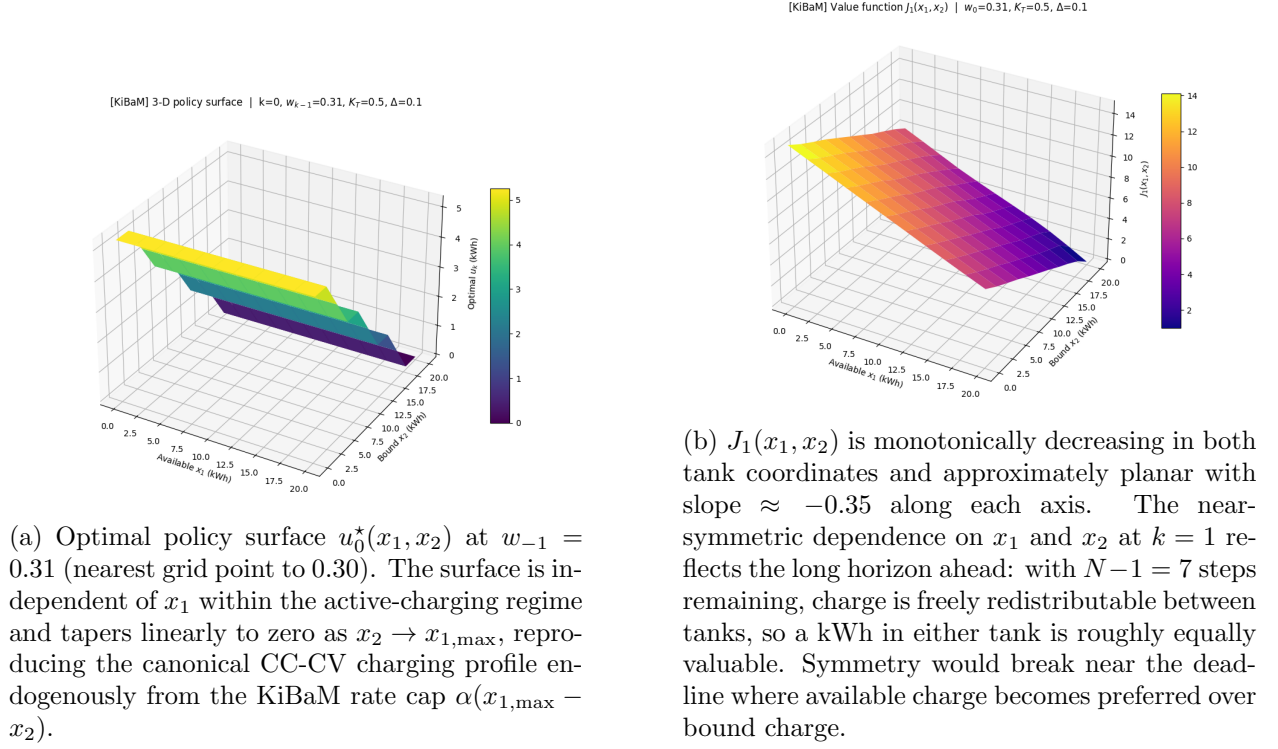


Figure 5: Three-dimensional views of the KiBaM solution.

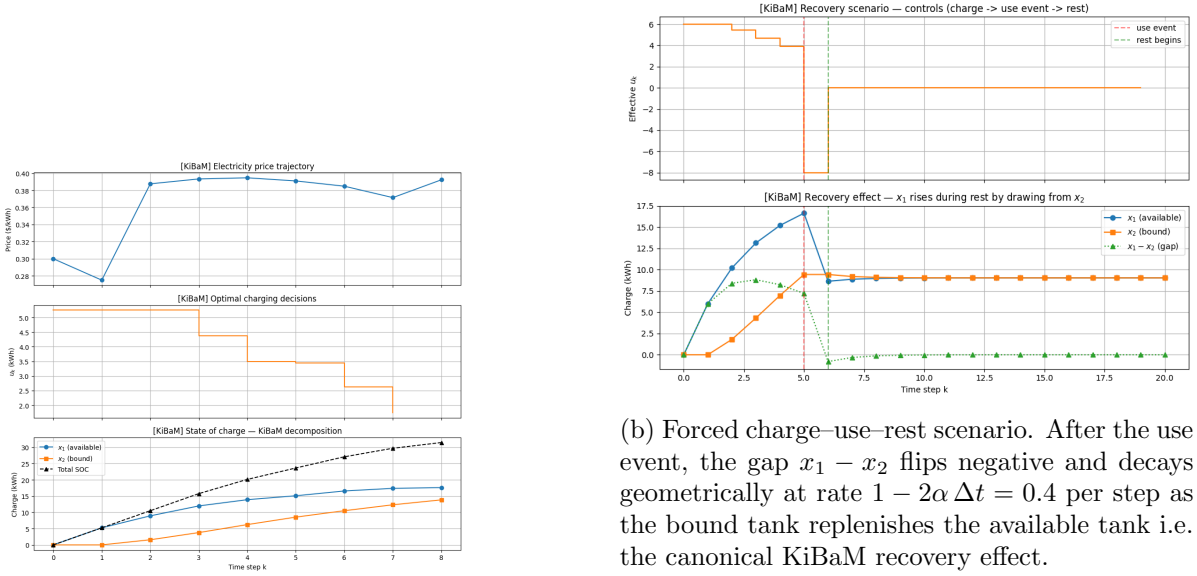


Figure 6: KiBaM dynamics in operation.

7.3 Comparison and Sensitivity

Table 2 reports the optimal initial control u_0^* and expected total cost J_0^* at the representative case for both models. The full 7×3 sweep over (K_T, Δ) for both models appears in Table 3.

Model	u_0^* (kWh)	J_0^* (\$)
Ideal	7.00	12.64
KiBaM	5.25	14.61

Table 2: Optimal initial control and expected total cost at the representative case ($K_T = 0.5$, $\Delta = 0.10$).

K_T	Δ	Ideal		KiBaM	
		u_0^*	J_0^*	u_0^*	J_0^*
0.1	0.05	0.00	4.00	0.00	4.00
0.1	0.10	0.00	4.00	0.00	4.00
0.1	0.20	0.00	4.00	0.00	4.00
0.2	0.05	0.00	7.94	0.00	7.96
0.2	0.10	0.00	8.00	0.00	8.00
0.2	0.20	0.00	8.00	0.00	8.00
0.3	0.05	0.00	11.25	0.00	11.51
0.3	0.10	0.00	11.08	0.00	11.42
0.3	0.20	0.00	11.45	0.00	11.59
0.4	0.05	7.00	12.12	5.25	13.33
0.4	0.10	7.00	12.64	5.25	13.61
0.4	0.20	7.00	13.31	5.25	14.09
0.5	0.05	7.00	12.12	5.25	14.25
0.5	0.10	7.00	12.64	5.25	14.61
0.5	0.20	7.00	14.45	5.25	15.98
0.7	0.05	7.00	12.12	5.25	16.00
0.7	0.10	7.00	12.64	5.25	16.36
0.7	0.20	7.00	14.45	5.25	17.72
1.0	0.05	7.00	12.13	5.25	18.62
1.0	0.10	7.00	12.64	5.25	18.98
1.0	0.20	7.00	14.45	5.25	20.35

Table 3: Sensitivity comparison between Ideal and KiBaM models across all tested K_T and Δ .

Sensitivity to K_T . K_T is the per-kWh cost of leaving the charging session with unmet demand, and it acts as the price threshold against which the spot price w_k is compared in the optimal policy. Increasing K_T makes shortfall more expensive, so the policy charges more aggressively at marginal prices and u_0^* rises. In the scenario of a large value K_T , the optimal policy charges at full feasible rate regardless of price and in the case of a smaller value K_T , the policy may deliberately tolerate shortfall when prices are unfavorable.

Sensitivity to Δ . Δ controls the conditional variance of the price process without changing its mean. Because the stage cost $g_k = w_k u_k$ is linear in w_k , the certainty-equivalent principle (Section 3) implies that u_0^* should be nearly insensitive to Δ . The sensitivity table confirms this empirically: u_0^* varies negligibly across Δ for fixed K_T , while J_0^* varies modestly through second-order curvature in the cost-to-go.

8 Conclusion

The ideal-battery DP yields a clean bang-bang threshold policy with the threshold determined by the terminal penalty K_T , consistent with the linear structure of the stage cost. The KiBaM extension preserves the threshold structure on the price axis but introduces state-dependent rate limits that produce CC-CV-like tapering endogenously. The recovery effect, visible in the forced charge–use–rest experiment, is the qualitative phenomenon that distinguishes KiBaM from the ideal model and could not be reproduced by the single-tank formulation at any choice of parameters.

Quantitatively, the KiBaM model produces a higher expected cost than the ideal model at the representative case ($J_0^* = 14.61$ vs. 12.64). The EV departs with only ~ 31 of 40 kWh under the optimal KiBaM policy because the rate cap $\alpha(x_{1,\max} - x_{2,k})$ binds before the deadline, whereas the ideal model achieves full charge with two periods to spare. The policy is also bang-bang in K_T across the sweep, with the transition occurring at $K_T \approx \mathbb{E}[w_0]$. Finally, as the stage cost is linear in w_k , the certainty-equivalent principle implies u_0^* is invariant to Δ at fixed K_T , while J_0^* retains second-order sensitivity through the curvature of the cost-to-go, as confirmed in Table 3.

Limitations: The price process is uniform with constant parameters K and Δ , so it captures neither seasonality (off-peak vs. peak hours) nor calendar effects (weekends, holidays). The KiBaM captures rate-capacity and recovery effects but ignores temperature, voltage limits, and capacity fade. All of which matter on longer timescales than a single charging session. Both models also assume unidirectional charging ($u_k \geq 0$) and vehicle-to-grid operation is not considered.

References

- [1] D. P. Bertsekas, *Dynamic Programming and Optimal Control, Vol. I*, 4th ed., Athena Scientific, 2017.
- [2] J. F. Manwell and J. G. McGowan, “Lead acid battery storage model for hybrid energy systems,” *Solar Energy*, vol. 50, no. 5, pp. 399–405, 1993.